



Muscle histology changes in first-lactation dairy cows from pre- to postpartum

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ABSTRACT

To support the nutritional demands of fetal growth and the onset of milk production, dairy cattle mobilize body tissues; and though this has been established in multiparous animals, the extent of this mobilization is less understood in primigravid dairy cattle. The objectives of this study were to characterize the *longissimus dorsi* muscle depth (LDD) and extent of LDD and backfat mobilization, as well as to characterize the muscle histological differences in pre- and postpartum primigravid dairy animals with varying muscle reserves as they transition into their first lactation. This prospective cohort study enrolled 21 heifers that were post hoc divided into 2 groups: high muscle (HM; LDD > 5.28 cm, n = 10) and low muscle (LM; LDD ≤ 5.28 cm, n = 11), based on the day of enrollment of the study, 5 wk before calving. Weekly measurements of LDD, BW, BCS, and insulin were recorded from enrollment until 28 DIM. Milk weights and DMI were recorded daily, and milk components were measured weekly. Biopsies of the *longissimus dorsi* muscle were collected ~3 wk before and after parturition. Biopsy samples were sectioned and analyzed for myofiber type profile and fiber type-specific cross-sectional area. The statistical model for repeated measures included the fixed effects of muscle group, time, their interaction, and the random effect of cow within group. The statistical model for single measurements per cow included the fixed effect of muscle group. Animals in the LM group produced 2.3 kg/d more milk compared with animals in the HM group, with no differences in component yields or ECM yield between groups across the first 4 wk in lactation. From 1 to 4 wk in lactation, HM cows lost 20 kg less BW than LM cows during the same time. The mean cross-sectional area of type I, IIA, or IIX myofibers was not different between groups in either the pre- or postpartum stage. Animals in the HM group had a greater proportion of type IIX fibers

than LM cows, which had more type IIA fibers than HM cows. Animals with lower LDD before calving lost more BW through 4 wk in lactation but produced more milk compared with animals having higher LDD. Prepartum muscle depth affected BW loss, milk yield, and muscle fiber type in primigravid animals. These results indicate that greater abundance of glycolytic myofibers may limit milk yield at the onset of lactation due to greater glucose demands by muscle, suggesting that muscle amount and myofiber profile may be associated with differences in milk yield.

Key words: *longissimus dorsi*, muscle depletion, muscle accretion, histology, primiparous

INTRODUCTION

The transition period is an especially critical time for first-lactation dairy animals as they are allocating nutrients to growth, fetal growth, mammary development, and eventual milk production. Transition cattle mobilize body reserves to account for the decreased DMI leading up to parturition and the negative energy and protein balance that follows (Bauman and Currie, 1980). Cows mobilize both adipose and muscle reserves to meet these increased energy and protein demands (McCabe and Boerman, 2020). The amount of tissue mobilized is highly variable by cow, but issues tend to arise when animals mobilize excessive tissue, which may be associated with adverse health events (Mann, 2022). Body composition and muscle reserves have been studied in multiparous cows but much less so in primigravid animals. Dairy cattle will begin to mobilize muscle reserves before calving to account for the prepartum AA deficiencies that occur and will continue to mobilize muscle through ~30 to 60 DIM (van der Drift et al., 2012; Mann et al., 2016; Hanno et al., 2025). Due to primigravid dairy cows continuing to allocate nutrients for growth, the extent and timing of their muscle mobilization relative to parturition is unknown.

Skeletal muscle is the largest internal organ and is used for the adaptation to lactation as well as for maintaining glucose homeostasis. Muscle turnover, defined as the

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

balance between protein synthesis and breakdown (Sadri et al., 2023), occurs within muscle that is composed of 3 primary fiber types: type I, IIA, and IIX (Picard and Gagaoua, 2020). These fibers differ in their metabolic characteristics, relying on either oxidative or glycolytic pathways to generate energy for contraction. Type I myofibers are slow-twitch oxidative fibers, type IIA are intermediate fast-twitch fibers, which are both oxidative and glycolytic, and type IIX are fast-twitch glycolytic fibers (Picard and Gagaoua, 2020). Muscle fiber typing has been less researched in dairy cattle, whereas extensive work has been done with beef cattle due to its correlation with tenderness and other meat quality traits. Compared with beef cattle, dairy cattle exhibit higher prevalence of oxidative type I myofibers in *semitendinosus* muscle because they are secretory-type animals and will partition more energy toward milk production instead of increasing myofiber size for meat production (Bellmann et al., 2004; Wicks et al., 2019; Wang et al., 2024). The changes of the transition period, which prepare the cow for increased milk production, may also influence muscle fiber switching due to factors such as age, activity, diet, or hormone concentrations (Picard and Gagaoua, 2020).

Whereas several studies have evaluated the extent of muscle mobilization in multiparous dairy cattle (i.e., van der Drift et al., 2012; McCabe et al., 2021; Gouveia et al., 2024), the same research has not been conducted in primigravid animals. Therefore, the objectives of this study were to characterize the extent of muscle depth and tissue mobilization in primigravid animals from late gestation through early lactation and then to characterize the histological differences in muscle fiber type and fiber type-specific cross-sectional area (CSA) in pre- and postpartum primigravid dairy cattle that exhibited varying amounts of muscle reserves. We hypothesized that animals in the high muscle (HM) group with greater *longissimus dorsi* depth (LDD) prepartum would display greater myofiber size in the pre- and postpartum periods and would have increased milk production compared with animals in the low muscle (LM) group with decreased LDD. We further hypothesized that the cows with greater LDD would have a greater proportion of slow-twitch type I muscle fibers because animals that prioritize nutrients for milk production tend to have increased oxidative fibers.

MATERIALS AND METHODS

Animals

For this prospective cohort study, all protocols and procedures were approved by the Purdue University Institutional Animal Care and Use Committee (protocol #0324002488) before the start of the study. Data were collected from primigravid Holstein dairy animals

(n = 21) from 5 wk before expected calving until 4 wk in lactation. All available primigravid heifers at the Purdue Dairy were enrolled that calved between August 28 and November 15, 2024, with no exclusion criteria used. These primigravid animals averaged 609 ± 72 kg (mean $\pm$ SD) BW and 675 ± 22 d of age at study enrollment. The heifers were split, post hoc, into HM (5.80 ± 0.12 ; n = 10) and LM (5.19 ± 0.12 ; n = 11) groups based on LDD on day of enrollment in the study. Cows with muscle depth greater than 5.28 cm were classified as HM and cows with 5.28 cm or less were classified as LM. The muscle group cutoff was determined post hoc after all -5 wk ultrasound images had been captured and measured to achieve near equal distribution between groups. Within the population, a natural separation in muscle depth occurred at ~ 5.28 cm, consequently resulting in unequal group distribution. The SAS (version 9.4; SAS Institute, Cary, NC) was used for a post hoc sample size estimation. The number of cows was calculated to detect a minimum of $1,080 \pm 668 \mu\text{m}^2$ difference in type I CSA between groups, assuming a power of 0.8 and a 2-tailed α of 0.05, 16 animals were needed (using data adapted from Bellmann et al., 2004).

Throughout the study, animals were fed common diets for both the nonlactating and lactating periods (Table 1) once a day between 0500 and 0600 h, with access to water at each stall, and milked twice per day at 0500 and 1600 h. Cows were housed in a tiestall barn on mattresses bedded with sawdust for the duration of the study. Animals were brought into the tiestall from pasture 7 wk before expected calving and remained in the tiestall until 4 wk after calving. During the nonlactating period, heifers were given ~ 30 min of exercise in an outdoor pasture each morning, and after beginning lactation, the animals had twice-daily exercise when walking to and from the milking parlor (~ 250 m).

Data Collection

BW and BCS. From 5 wk before expected calving to 4 wk in lactation, cows were weighed weekly on the same day of the week for a total of 9 time points per cow of -5 , -4 , -3 , -2 , -1 , 1 , 2 , 3 , and 4 wk relative to calving. Before the start of afternoon milking, prepartum heifers were weighed, and fresh animals were weighed after afternoon milking as they exited the milking parlor, using a stationary scale (IQ+355-2A, Rice Lake Weighing Systems, Muskego, WI). While animals were weighed, 3 trained observers recorded a BCS using a 0 to 5 point scale with 0.25 point increments (Wildman et al., 1982), which was averaged for a weekly BCS for each animal at the same time points as BW.

Ultrasound. From 5 wk before expected calving until 4 wk in lactation, ultrasound images were collected every 7

Table 1. Ingredient and nutrition composition of the nonlactating and lactating diet (values are $\pm$ SD)

Composition	Nonlactating diet	Lactating diet
Ingredients, g/100 g diet DM		
Corn silage	28.6	39.3
Alfalfa silage	—	14.7
Triticale silage	50.0	3.6
Ground corn	—	14.1
Soybean meal	6.6	7.9
Soyhull pellets	13.6	6.9
Blood meal	—	1.24
Amino Plus ¹	—	4.1
QLF 68/38 ²	—	4.7
Dry cow premix ³	1.1	—
Lactating cow premix ⁴	—	2.4
Fat and protein blend ⁵	—	1.2
Nutrients ⁶		
DM, % as fed	48.9 $\pm$ 2.9	49.9 $\pm$ 2.0
CP	11.5 $\pm$ 1.3	16.7 $\pm$ 0.14
ADF	35.4 $\pm$ 1.5	19.3 $\pm$ 1.8
aNDF ⁷	52.7 $\pm$ 0.6	27.3 $\pm$ 1.1
Starch	12.3 $\pm$ 0.6	27.6 $\pm$ 1.1
Ash	7.03 $\pm$ 0.6	7.12 $\pm$ 1.1
NEL, Mcal/kg	1.44 $\pm$ 0.02	1.66 $\pm$ 0.05

¹Ag Processing, Omaha, NE.²Quality Liquid Feeds, Dodgeville, WI.³Dry cow premix contained 29.5% calcium carbonate, 18.7% magnesium sulfate, 18.7% magnesium oxide, 18.1% vitamin E 20,000 IU, 6.8% salt, 4.1% Akey Dairy, 2.8% Zinpro Availa, 1.3% selenium (0.06% inorganic selenium), 0.22% Purdue ADE.⁴Lactating cow premix contained 22.7% calcium carbonate, 21.1% sodium carbonate, 11.6% salt, 11.3% CalMin (Celtic Sea Minerals, Ireland), 7.9% DCAD Plus (Arm & Hammer), 7.5% Biofos, 7.1% urea, 6.1% Mag-Ox, 1.9% Diamond V XPC (Diamond V, Cedar Rapids, IA), 1.2% Dairy PMX, 1.0% Availa-4, 0.4% vitamins A, D, E, 0.2% selenium.⁵Fat and protein blend contained 51.8% Spectrum fusion (Perdue, Salisbury, MD), 30.5% Megalac (Volac Wilmar Feed Ingredients, Liverpool, UK), 8.6% Alimet (Novus, Saint Charles, MO), 4.4% ETX-5 (Impextraco, Antwerp, Belgium), 4.3% Smartamine M (Adiesseo, Alpharetta, GA), 0.4% Agolin (Agolin, Switzerland).⁶Analyzed from 2 samples each of the nonlactating and lactating diets.⁷Ash-free NDF analyzed with amylase sodium sulfite by Cumberland Valley Analytical Services (Waynesboro, PA).

d after the afternoon milking for a total of 10 time points per cow (similar to the BW time points). Images were taken of the *longissimus dorsi* muscle above the 12th intercostal space on the right side of each cow and ~2.5 cm distal to the transversal process of the vertebral column by 1 trained researcher, with 3 images captured per cow per time point. An ExaGo ultrasound (imvImaging, Rochester, MN) was used with a 130 mm linear probe to capture images, utilizing the method from Megahed et al. (2019) and McCabe et al. (2021) with a different brand of ultrasound.

Blood Sampling and Analysis. Blood samples were collected weekly within 1 h of feeding from the coccygeal vessels using a 21-gauge needle (McKesson, Irving, TX) into 10-mL potassium EDTA tubes (Becton, Dickinson, and Company, Franklin Lakes, NJ). The blood samples were centrifuged at $4,000 \times g$ at 25°C for 15 min. Sepa-

rated plasma was aliquoted into 2 1.5-mL microcentrifuge tubes and stored at –20°C until further analysis. Blood plasma samples for –4, –3, –2, –1, 1, 2, 3, and 4 wk relative to calving were analyzed for insulin concentrations using insulin kits from ALPCO (80-INSPO-EO1, Salem, NH). Intraplate CV were 11.8% and interplate CV was 18.0%.

Colostrum and Calf Sampling. Calves were weighed within 1 h of being born on the same stationary scale as the cows before being fed colostrum. Colostrum was collected from the cow at the first milking after parturition and was weighed (PCE-HS 50N, PCE Instruments, Jupiter, FL) before Brix analysis for IgG content.

Milk Yield, Sampling, and Analysis. Cows were milked twice daily at 0500 and 1600 h, daily milk yield was collected using Afimilk (Kibbutz Afikim, Israel). Components were collected weekly at 2 consecutive milkings for the first 4 wk of lactation. Milk samples were analyzed for fat, protein, lactose, solids, and MUN using Fourier-transform infrared spectroscopy (MilkoScan 7 RM, Foss, Hillerød, Denmark) at Dairy One (Ithaca, NY). Weekly concentrations of components were calculated by multiplying the concentration values from the milk test by the milk yield at that milking.

Feed Intake. Cows were fed ad libitum TMR daily for the duration of the study. This was calculated by weighing daily offerings of feed and refusals and feeding for 110% of the previous day's intake. The DM of the forages was analyzed weekly with a Koster tester, and the diet formulation was adjusted accordingly. Using weekly DM data from TMR tracker, daily DMI was calculated. Feed samples were taken monthly and analyzed by Cumberland Valley Analytical Services (Waynesboro, PA) to determine DM, CP (method 990.03; AOAC International, 2000), NDF (Van Soest et al., 1991), ADF (method 973.18; AOAC International, 2000), and starch (Hall, 2009) of the TMR samples.

Muscle Biopsies. Muscle biopsies of the *longissimus dorsi* were collected ~3 wk (19 ± 3 d) before expected calving and ~3 wk (19 ± 2 d) in lactation using a 5 mm Bergstrom biopsy needle. A portion of each biopsy was oriented longitudinally on plastic supports, covered with optimal cutting temperature (OCT) compound (Fisher Scientific, 23–730–571), and rapidly frozen in isopentane cooled on liquid nitrogen. The OCT-embedded muscle biopsy samples were stored at –80°C until further analysis.

Muscle Fiber Size and Type. Muscle tissue cross-sections (10 μ m) were cut from OCT-embedded muscle biopsy samples in a cryostat (Leica CM1950) at –20°C and adhered to SuperFrost Plus slides (Fisher Scientific, 12–550–15). Tissue sections were air-dried at room temperature, encircled with a hydrophobic PAP pen (Vector, H-4000), and then blocked for 1 h at room temperature

in 10% goat serum (Invitrogen, 10000C) prepared in PBS. Muscle cross-sections were then immunostained overnight at 4°C in blocking buffer containing a cocktail of primary antibodies to comprehensively assess the fiber type composition of adult bovine muscle tissue, as described by Song et al. (2020) with slight modifications. Briefly, the primary antibody cocktail included antibodies against laminin 1 + 2 (Abcam, ab7463), slow-twitch myosin heavy chain (MyHC) type I (Developmental Studies Hybridoma Bank [DSHB], BA-D5, mouse IgG2b 1:100), fast-twitch MyHC type IIX (DSHB, 6H1, mouse IgM 1:10), and all MyHC isoforms except for IIX (DSHB, BF-35c, mouse IgG1 1:100). This allowed for simultaneous staining of all the 3 major adult bovine MyHC isoforms on a single tissue section, enabling identification of 3 pure muscle fiber types (I, IIA, and IIX fibers; Figure 1). During initial method optimization, the reactivity of each individual primary antibody was tested in isolation when compared with application in cocktail format on serial sections of a representative bovine muscle biopsy sample. The reactivity of these primary antibodies was found to be highly similar when applied in cocktail format as when applied individually, consistent with the published findings of Song et al. (2020). Following overnight incubation, primary antibodies were detected with isotype-specific Alexa Fluor conjugated secondary antibodies (1:500 in PBS), including Goat Anti-Mouse IgG1 Alexa Fluor 488 (Invitrogen, A-21121; to detect BF-35), Goat Anti-Mouse IgG2b Alexa Fluor 555 (Invitrogen, A-21147; to detect BA-D5), Goat Anti-Mouse IgM Alexa Fluor 647 (Invitrogen, A-21238; to detect 6H1), and Goat Anti-Rabbit IgG (H⁺L) Alexa Fluor 350 (Invitrogen, A-11046; to detect ab7463). Slides were mounted with coverslips using Mowiol Fluorescence Mounting Medium (Sigma-Aldrich). Stitched mosaic

images spanning the entire cross-section of fluorescent stained samples were imaged with an automated microscope equipped with a motorized stage operating in an upright configuration using a 10 × Plan Fluorite objective (Echo Revolution). The percentage of muscle fiber type composition and fiber type-specific myofiber CSA were manually counted and measured using ImageJ software (version 1.54i; NIH, Bethesda, MD) using the freehand selection tool. Myofibers were identified as type I (BA-D5^{pos}BF-35^{pos} cells), type IIA (BA-D5^{neg}BF-35^{pos} cells), or type IIX (6H1^{pos}BF-35^{neg} cells). To calculate percentage fiber type profile, the number of myofibers of each type was expressed as a proportion of the total number of myofibers observed and counted. All correctly oriented, for example, cross-sectional muscle fibers throughout the entire muscle biopsy cross-section were counted and measured (n = 37). The average number of total muscle fibers quantified per sample was 357.5 ± 28.0, which is within the range of other published work (Greenwood et al., 2009; Ebarb et al., 2017).

Data Analysis

Ultrasound images were used to measure back fat depth (BFD) and LDD using ImageJ software. The BFD and LDD values were averaged across the 3 captured images per cow per time point. The average CV of LDD and BFD for all time points was 4.3% and 12.6%, respectively. To evaluate the change in different stages of the transition period, measurements from the beginning and end of each stage (parturition from −5 to −1 wk and postpartum from 1 to 4 wk) were used to calculate absolute change in BFD and LDD across transition stages. Parturition mobilization of muscle and backfat was defined as muscle or backfat at −5 wk minus muscle or backfat at −1 wk,

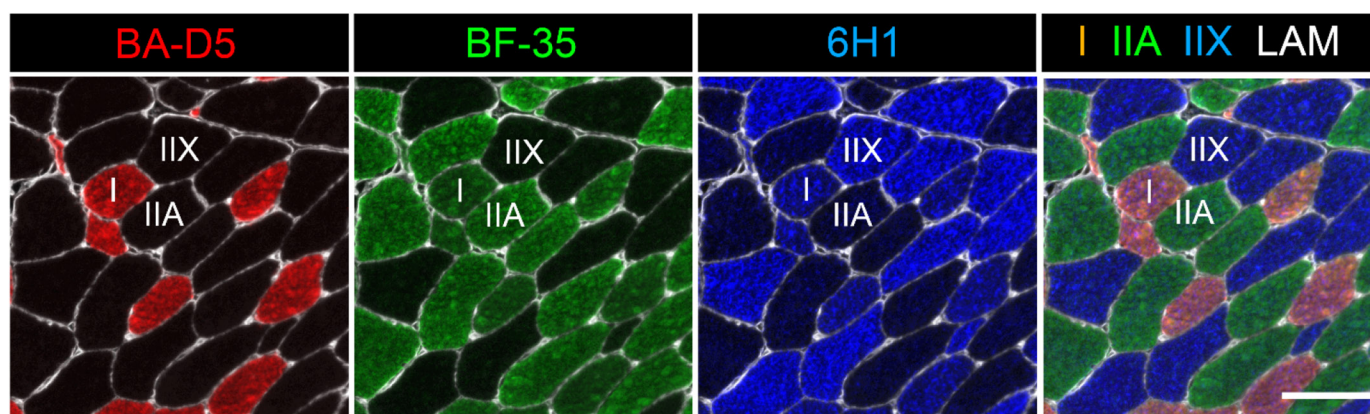


Figure 1. Representative image of staining of *longissimus dorsi* muscle samples from primigravid Holstein heifers. The different primary antibodies are listed with BA-D5 staining type I myofibers, BF-35 staining both type I and type IIA, and 6H1 staining type IIX fibers. A primary antibody against laminin was used to define cell borders. Scale bar is 100 μ m.

respectively. The postpartum mobilization of muscle and backfat was defined as muscle or backfat at 1 wk minus muscle or backfat at 4 wk.

Statistical Analysis

Data were analyzed using the MIXED procedure available in SAS (version 9.4; SAS Institute, Cary, NC). The Shapiro–Wilk test was used to test normality, and all data were normally distributed. The model for repeated measures is described as

$$Y_{ijkl} = \mu + M_i + T_j + MT_{ij} + C(M_i)_k + e_{ijkl},$$

where μ is the overall mean, M_i represents the fixed effect of muscle group (i = HM vs. LM), T_j represents the fixed effect of time (j = −5, −4, −3, −2, −1, 1, 2, 3, 4 wk relative to calving, or DIM or stage of lactation [pre- vs. postpartum]), MT_{ij} represents the interaction of muscle group and time, and $C(M_i)_k$ is the random effect of cow nested within muscle group. Repeated measures with cow as the subject were used with the first-order autoregressive covariance structure, which minimized Bayesian information criterion and Akaike information criterion compared with other covariate structures evaluated. Kenward–Roger adjustment was used due to small sample size with unequal group distribution. For

nonrepeated measures, the fixed effect of muscle group and the random effect of cow nested within muscle group were used in the model. Group, time, and interaction terms at $P < 0.05$ were considered significant, and $0.05 \leq P \leq 0.15$ were considered a tendency. The LSM and SEM were reported for muscle groups unless otherwise stated.

RESULTS

BW and Tissue Reserves

During the prepartum period, both HM and LM heifers gained an average of 44.5 ± 2.7 kg (Table 2), with no difference between muscle groups ($P = 0.58$). Postpartum HM cows lost 26.4 ± 6.7 kg through 4 wk in lactation, whereas LM cows lost 45.8 ± 6.4 kg ($P = 0.05$) in the same time frame.

Heifers in the HM group had more muscle at −5 wk than heifers in the LM group (6.22 ± 0.19 vs. 5.44 ± 0.18 cm; $P < 0.01$, Figure 2). At wk 1 in lactation, the HM cows had more muscle than the LM cows at the same time point (5.61 ± 0.19 cm vs. 5.00 ± 0.18 cm; $P = 0.02$). By wk 4, the HM cows tended to have more muscle than the LM animals (5.16 ± 0.18 cm vs. 4.62 ± 0.19 cm; $P = 0.05$). Over the course of the study, both the HM and LM cows mobilized 0.19 ± 0.07 cm of back fat ($P = 0.68$).

Table 2. Associations between prepartum skeletal muscle reserves and DMI, average muscle depth, backfat depth, BW, BCS, plasma insulin, and change of muscle depth, backfat depth, and BW

Variable	Muscle group ¹		SEM ³	P-value ²
	HM	LM		
Prepartum (−5 to −1 wk)				
BW (wk −5), kg	624.8	623.3	15.7	0.97
BCS (wk −5)	3.51	3.29	0.10	0.17
Backfat depth (wk −5), cm	0.74	0.63	0.06	0.22
Muscle depth, cm	6.22	5.44	0.18	<0.01
BW change, kg	42.6	46.4	4.84	0.58
Backfat change, cm	−0.05	0.003	0.06	0.53
LDD change, cm	−0.44	−0.03	0.22	0.19
Insulin, ng/mL	0.45	0.32	0.05	0.07
Postpartum (1 to 4 wk)				
BW (wk 4), kg	574.6	550.0	15.4	0.57
BCS (wk 4)	3.46	3.30	0.10	0.73
Backfat depth (wk 4), cm	0.55	0.45	0.04	0.09
Muscle depth (wk 4), cm	5.16	4.62	0.20	0.06
BW change, kg	−26.4	−45.8	6.70	0.05
Backfat change, cm	−0.11	−0.13	0.05	0.81
LDD change, cm	−0.46	−0.33	0.21	0.67
Insulin, ng/mL	0.32	0.24	0.04	0.11
Overall (−5 to 4 wk)				
BW change	−50.2	−73.3	8.39	0.06
Backfat change, cm	−0.20	−0.16	0.07	0.68
LDD change, cm	−1.07	−0.76	0.30	0.46

¹Cows were assigned to muscle group based on *longissimus dorsi* depth (LDD) at −5 wk before expected calving. High-muscle LDD was >5.28 cm ($n = 10$). Low-muscle LDD was ≤ 5.28 cm ($n = 11$).

²P-values associated with muscle group effect (HM vs. LM).

³SEM determined by the largest group SEM.

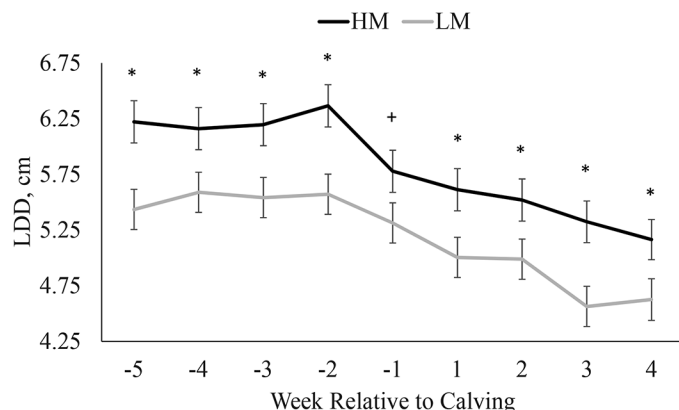


Figure 2. *Longissimus dorsi* depth (LDD) collected from ultrasound images of primigravid Holstein heifers from -5 to 4 wk relative to calving. Cows were divided into groups; high muscle = HM; LDD >5.28 cm or low muscle = LM; LDD ≤5.28 cm at -5 wk relative to calving. Error bars indicate SEM. * indicates $P < 0.05$, and + indicates $P < 0.10$.

DMI

In the prepartum period, the HM group consumed 0.56 ± 0.09 kg/d less DM than LM animals ($P < 0.01$, Table 3), whereas in the postpartum period, HM animals consumed 0.35 ± 0.12 kg/d more DM compared with LM animals ($P = 0.03$).

Milk Production

The HM cows produced 2.3 ± 0.74 kg/d less milk compared with LM cows ($P = 0.02$). Cows in the HM group tended to have a higher concentration of fat in their milk ($5.43\% \pm 0.17\%$ vs. $5.03\% \pm 0.16\%$; $P = 0.11$) but not

protein ($3.47\% \pm 0.13\%$ vs. $3.27\% \pm 0.12\%$; $P = 0.27$) compared with LM cows, however, muscle group did not affect yields of milk protein and fat ($P = 0.64$ and $P = 0.65$, respectively). Energy-corrected milk did not differ between groups ($P = 0.52$).

Plasma Insulin

The HM animals tended to have increased insulin concentrations in the prepartum and postpartum periods compared with LM animals ($P = 0.07$, $P = 0.11$, respectively).

Muscle Histology

The percentage of type I myofibers in biopsy samples of the *longissimus dorsi* did not differ by muscle group ($P = 0.53$) or stage ($P = 0.95$, Table 4). However, HM cows had lower percentages of type IIA myofibers compared with LM cows ($35.1\% \pm 1.37\%$ vs. $39.7\% \pm 1.19\%$; $P = 0.02$) with no effect of stage of lactation. Cows in the HM group had a greater percentage of type IIX myofibers than the LM group ($36.4\% \pm 1.68\%$ vs. $30.5\% \pm 1.46\%$; $P = 0.01$). There were no differences in the proportion of type IIX myofibers between the pre- and postpartum periods ($P = 0.40$). No muscle group by stage interactions were observed for the relative proportions of type I, IIA, or IIX myofibers. Surprisingly, animals had similar mean myofiber CSA of all muscle fiber types across muscle groups and stages of lactation. Overall, type I ($2,563 \pm 173.6$) myofibers had the smallest mean CSA, followed by IIA ($4,832 \pm 395.3$) and then IIX ($5,497 \pm 439.3$).

Table 3. Associations between prepartum skeletal muscle reserves and milk yield, ECM, and milk components

Variable	Muscle group ¹			P-value ²		
	HM	LM	SEM ³	Muscle group	Time	Muscle group × time
DMI						
Prepartum, kg/d	8.9	9.5	0.09	<0.01	0.01	0.84
Postpartum, kg/d	16.0	15.6	0.12	0.03	<0.01	0.12
Milk yield, kg/d	24.6	26.9	0.74	0.02	<0.01	0.97
28-d milk yield, kg	685.2	756.6	35.0	0.16	—	—
ECM, kg/d	31.3	33.0	1.88	0.52	0.12	0.60
Milk fat, kg/d	1.33	1.37	0.09	0.74	0.22	0.46
Milk protein, kg/d	0.84	0.87	0.04	0.51	0.12	0.89
Milk fat, %	5.43	5.03	0.17	0.11	0.01	0.51
Milk protein, %	3.47	3.27	0.13	0.27	<0.01	0.73
MUN, mg/dL	12.7	13.1	0.42	0.51	0.93	0.12
SCS ⁴	2.11	2.13	0.20	0.95	<0.01	0.97

¹Cows were assigned to muscle group based on *longissimus dorsi* depth (LDD) at -5 wk before expected calving. High-muscle LDD was >5.28 cm ($n = 10$). Low-muscle LDD was ≤5.28 cm ($n = 11$).

²P-values associated with muscle group (HM vs LM), time (wk), and their interaction.

³SEM determined by the largest group SEM.

⁴The SCC was converted to SCS using $\log_2 \left(\frac{\text{SCC}}{100} \right) + 3$, as described by (Wiggans and Shook, 1987).

Table 4. Associations between prepartum skeletal muscle reserves¹ and myofiber types and cross-sectional area of myofibers collected from biopsies –3 wk (prepartum) and 3 wk (postpartum) relative to parturition

Variable	Prepartum		Postpartum		SEM ³	P-value ²		
	HM	LM	HM	LM		Muscle group	Stage	Muscle group × Stage
Proportion of myofiber								
Type I	29.5	28.6	27.4	31.0	2.27	0.53	0.95	0.30
Type IIA	35.3	41.5	34.9	37.9	1.93	0.02	0.27	0.39
Type IIX	35.2	29.9	37.7	31.1	2.37	0.01	0.40	0.76
Myofiber size								
Type I, μm^2	2,369	2,623	2,626	2,638	262	0.59	0.58	0.63
Type IIA, μm^2	4,452	5,011	4,733	5,136	597	0.40	0.72	0.89
Type IIX, μm^2	5,219	5,736	5,436	5,598	663	0.59	0.95	0.78

¹Cows were assigned to muscle group based on *longissimus dorsi* depth (LDD) at –5 wk before expected calving. High-muscle LDD was >5.28 cm (n = 10). Low-muscle LDD was ≤5.28 cm (n = 11).

²P-values associated with group effect (HM vs. LM), time effect (wk), and their interaction.

³SEM determined by the largest group SEM.

DISCUSSION

Animals in this study mobilized muscle and fat in similar patterns regardless of muscle group during their transition into lactation. Animals with more muscle depth mobilized more LDD than those with less LDD and also lost less BW through 4 wk in lactation despite cows in both groups weighing the same prepartum. There were no differences in ECM between groups, as LM cows made more milk yield, but HM cows had higher component concentrations. The cows that were in the HM group had an increased number of type IIX muscle fibers compared with animals in the LM group that had increased numbers of type IIA fibers.

Both the HM and LM cows maintained their LDD in the prepartum period and mobilized muscle in the postpartum period. Previous studies have shown that multiparous animals with increased LDD will begin to mobilize muscle before calving, whereas multiparous animals with lesser LDD will continue to accrete muscle before calving; however, both groups mobilize muscle post calving (McCabe et al., 2021; Gouveia et al. 2024). Another study found that there was no change in LDD in the prepartum period, whereas backfat increased in the same period (Westhoff et al. 2024). The previous studies used multiparous animals, whereas the current paper only looked at primigravid animals. This could explain the difference in the LDD patterns, due to the fact that immature animals are still allocating nutrients to growth. The primigravid animals in this study lost, on average, 8% of their muscle depth from parturition to 4 wk in milk, whereas multiparous animals within the same herd lost 20.5% of their muscle depth in the same time period (McCabe et al., 2021; Gouveia et al., 2024). This shows that primigravid animals are mobilizing less skeletal muscle, potentially due to the fact that they still

have to allocate nutrients to growth, whereas multiparous animals do not.

Some previous studies have also shown that the higher-muscled animals prepartum allocated nutrients toward the growth of the fetus and had calves with a higher birth weight (McCabe et al., 2021; Beckett et al., 2025). There was no difference in birth weight in calves in this trial, though the authors recognize that there was a small sample size (Table 5). This also supports the theory that primigravid animals allocate nutrients differently than mature animals and may retain more nutrients for their growth as opposed to allocating them for calf growth. Primigravid cows exhibit a decreased extent of negative energy balance compared with mature animals, showing lower BHB concentrations and lower milk yield. The lower milk yield requires less demand for nutrients, allowing these immature animals to allocate nutrients toward growth (Walter et al., 2022). Heifers are expected to calve for the first time between 21 and 24 mo. of age; however, at this point they are not fully mature and will not be until ~60 mo of age (Coffey et al., 2006; Wathes et al., 2007). Primigravid cows have a differing homeorhetic mechanism to assist with their metabolic partitioning (McArt et al., 2013). Primigravid cows are balancing demands for growth with milk production, allowing them to more readily shift their metabolic partitioning in response to health challenges (Bauman and Currie, 1980). This shift, as well as the decreased production, leaves primigravid animals less susceptible to metabolic issues.

Muscle fiber types vary in their size, color, and myosin chain isoforms (Schiaffino and Reggiani, 1994; Hoh, 2023). Immunohistochemistry is a newer technique used to stain and differentiate myofibers that has been validated for use with cattle muscle (Schiaffino 2018; Song et al., 2020). When using a cocktail of antibodies for staining, it is possible to identify the 3 types of myofibers present

Table 5. Means and SD for calf and colostrum variables collected from primigravid Holstein cows

Variable	Muscle group ¹	
	HM	LM
Calf weight, kg	41.6 ± 3.73	41.9 ± 4.14
Colostrum wt, kg	6.80 ± 3.02	8.25 ± 4.05
Brix, %	31.0 ± 2.64	28.4 ± 4.7
Proportion female calf	7 out of 10	9 out of 11

¹Cows were assigned to muscle group based on *longissimus dorsi* depth (LDD) at -5 wk before expected calving. High-muscle LDD was >5.28 cm (n = 10). Low-muscle LDD was ≤5.28 cm (n = 11).

in cattle: type I, IIA, and IIX. The staining procedure used allowed for all 3 types of myofibers to be visualized in this set of primigravid dairy cattle. Glycolytic muscle fibers, such as the type IIX fibers, use more glucose as they employ an anaerobic metabolic pathway for contraction (Picard and Gagaoua, 2020). This increased requirement for glucose by muscle tissue in early lactation may reduce available glucose to the mammary gland, leading to less glucose available for milk production. Animals in the HM group had an increased proportion of type IIX myofibers, which may have contributed to the decreased milk production in these animals.

Muscle CSA is associated with muscle fiber type, where oxidative muscle fibers are the smallest and intermediate and glycolytic muscle fibers are larger (Picard and Gagaoua, 2020). In cattle, muscle CSA and fiber type change as animals age, with older animals having larger CSA and greater percentages of type I and IIA as opposed to IIX myofibers. Muscle mass increases during postnatal life, generally not from an increase in the number of myofibers but from an increase in the relative size of the myofibers. Oxidative fibers tend to be smaller compared with glycolytic fibers in most muscles, although the CSA of different myofiber types depends on the specific muscle group (Picard and Gagaoua, 2020). Unexpectedly, we did not observe any differences in myofiber CSA for muscle group or stage in this set of primigravid animals. Increased ME intake leads to an increased frequency of type IIX myofibers, indicating that diet may affect myofiber type (Wicks et al., 2019). In the postpartum period, animals in both muscle groups tended to have higher proportions of type IIX myofibers, which may be linked to the increased DMI postpartum and because lactating diets are more energy dense than nonlactating diets. When evaluating the proportions of the varying types of myofibers, there is indication of myofiber switching occurring between the pre- and postpartum periods, which can be seen in the relative proportion of the type II myofibers changing between pre- and postpartum. There were no observed differences in fiber CSA between pre- and postpartum, which was unexpected.

The prepartum and postpartum differences in muscle histology are most likely due to the differing endocrine responses that the body uses to adapt to lactation, as skeletal muscle helps to regulate homeostasis and some hormones. Between the prepartum and postpartum periods there are differences in the circulating concentrations of both glucose and insulin in the body. Skeletal muscle is responsible for greater than 75% of the insulin-mediated glucose dispersal throughout the body (Feraco et al., 2021). Circulating glucose concentrations are elevated during the prepartum period due to limited mammary uptake. Following parturition, mammary glucose demand increases substantially to support lactation (Habel and Sundrum, 2022). The increased prepartum glucose concentrations are coupled with increased circulating insulin during this stage (De Koster and Opsomer, 2013). In contrast, the postpartum is characterized by reduced glucose and insulin concentrations, and insulin resistance in peripheral tissue is increased to partition nutrients to the mammary gland (De Koster and Opsomer, 2013). Animals in the HM group had higher insulin in both the pre- and postpartum periods compared with the LM animals. Although not evaluated in this study, insulin resistance has been shown to vary based on muscle type, with increased type IIX myofibers leading to increased insulin resistance in humans (Fisher et al., 2017). Future work would be needed to evaluate the insulin resistance and myofiber type interaction in primigravid dairy cattle.

There were limitations in this study because of the small sample size in this study, and therefore only the descriptive statistics were analyzed on the calf variables, although future work should consider the effects of muscle amount and mobilization on calf and colostrum variables. Taking biopsies is thought to be a representation of whole muscle; however, because the biopsy site shifted slightly to avoid potential scar tissue from the previous site, it is possible that myofiber distribution can differ across the same muscle. There may have been some effect due to the length of the study and lack of large cohorts of heifers that spread the study out over a longer period of time, which allowed for changes of forages in the diet, as well as some animals being in the barn for longer periods of time with differing environmental conditions. Ideally in the future, larger cohorts enrolled over a shorter time would reduce potential environmental effects. Further investigation into the extent of muscle mobilization in primigravid animals is needed. Following these animals for a whole lactation will allow us to understand the patterns of muscle accretion and mobilization and see if they are similar to multiparous cows. In addition, collecting muscle biopsies at multiple stages of lactation would allow for the identification of time points at which muscle fiber type switching may occur.

CONCLUSIONS

Primigravid dairy animals vary in their skeletal muscle reserves at 5 wk before calving. Animals with less muscle reserves showed a lower proportion of fast-twitch glycolytic IIX muscle fibers in the *longissimus dorsi* and produced more milk than higher-muscled animals, however, there was no difference in ECM yield between muscle groups. Primigravid animals mobilize tissue to a lesser extent than multiparous animals, most likely due to their requirements for growth and decreased milk production. There was also evidence of myofiber switching from type IIA to type IIX between the pre- and postpartum periods, showing that the transition to lactation causes fundamental changes within the body that affect milk production. Despite these changes, muscle fiber size did not differ in relation to skeletal muscle reserves or from pre- to postpartum.

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Nonstandard abbreviations used: aNDF = ash-free NDF; BFD = back fat depth; CSA = cross-sectional area; DSHB = Developmental Studies Hybridoma Bank; HM = high muscle; LDD = *longissimus dorsi* depth; LM = low muscle; MyHC = myosin heavy chain; OCT = optimal cutting temperature.

REFERENCES

- AOAC International. 2000. Official Methods of Analysis of AOAC International. 17th ed. Vol. 1 and 2. AOAC Int.
- Bauman, D. E., and W. B. Currie. 1980. Partitioning of nutrients during pregnancy and lactation: A review of mechanisms involving homeostasis and homeorhesis. *J. Dairy Sci.* 63:1514–1529. [https://doi.org/10.3168/jds.S0022-0302\(80\)83111-0](https://doi.org/10.3168/jds.S0022-0302(80)83111-0).
- Beckett, L. M., B. Gast, E. Tobolski, L. Jones, K. Gouveia, Y. Han-Hallett, T. Casey, and J. P. Boerman. 2025. Dam prepartum skeletal muscle reserves and supplementation with branched-chain volatile fatty acids during late gestation influence calf birth weight and calf muscle metabolic activity. *JDS Commun.* 6:293–298. <https://doi.org/10.3168/jdsc.2024-0581>.
- Bellmann, O., J. Wegner, F. Teuscher, F. Schneider, and K. Ender. 2004. Muscle characteristics and corresponding hormone concentrations in different types of cattle. *Livest. Prod. Sci.* 85:45–57. [https://doi.org/10.1016/S0301-6226\(03\)00121-0](https://doi.org/10.1016/S0301-6226(03)00121-0).
- Coffey, M. P., J. Hickey, and S. Brotherstone. 2006. Genetic aspects of growth of Holstein-Friesian dairy cows from birth to maturity. *J. Dairy Sci.* 89:322–329. [https://doi.org/10.3168/jds.S0022-0302\(06\)72097-5](https://doi.org/10.3168/jds.S0022-0302(06)72097-5).
- De Koster, J. D., and G. Opsomer. 2013. Insulin resistance in dairy cows. *Vet. Clin. North Am. Food Anim. Pract.* 29:299–322. <https://doi.org/10.1016/j.cvfa.2013.04.002>.
- Ebarb, S. M., K. J. Phelps, J. S. Drouillard, K. R. Maddock-Carlin, M. A. Vaughn, D. D. Burnett, J. A. Noel, C. L. Van Bibber-Krueger, C. B. Paulk, D. M. Grieger, and J. M. Gonzalez. 2017. Effects of anabolic implants and ractopamine-HCl on muscle fiber morphometrics, collagen solubility, and tenderness of beef *longissimus lumborum* steaks. *J. Anim. Sci.* 95:1219–1231. <https://doi.org/10.2527/jas.2016.1263>.
- Feraco, A., S. Gorini, A. Armani, E. Camajani, M. Rizzo, and M. Caprio. 2021. Exploring the role of skeletal muscle in insulin resistance: Lessons from cultured cells to animal models. *Int. J. Mol. Sci.* 22:9327. <https://doi.org/10.3390/ijms22179327>.
- Fisher, G., S. T. Windham, P. Griffin, J. L. Warren, B. A. Gower, and G. R. Hunter. 2017. Associations of human skeletal muscle fiber type and insulin sensitivity, blood lipids, and vascular hemodynamics in a cohort of premenopausal women. *Eur. J. Appl. Physiol.* 117:1413–1422. <https://doi.org/10.1007/s00421-017-3634-9>.
- Gouveia, K. M., L. M. Beckett, T. M. Casey, and J. P. Boerman. 2024. Production responses of multiparous dairy cattle with differing prepartum muscle reserves and supplementation of branched-chain volatile fatty acids. *J. Dairy Sci.* 107:11655–11668. <https://doi.org/10.3168/jds.2024-24915>.
- Greenwood, P. L., N. W. Tomkins, R. A. Hunter, P. G. Allingham, S. Harden, and G. S. Harper. 2009. Bovine myofiber characteristics are influenced by postweaning nutrition. *J. Anim. Sci.* 87:3114–3123. <https://doi.org/10.2527/jas.2009-1936>.
- Habel, J., and A. Sundrum. 2020. Mismatch of glucose allocation between different life functions in the transition period of dairy cows. *Animals (Basel)* 10:1028. <https://doi.org/10.3390/ani10061028>.
- Hall, M. B. 2009. Determination of starch, including Maltooligosaccharides, in animal feeds: Comparison of methods and a method recommended for AOAC Collaborative Study. *J. AOAC Int.* 92:42–49. <https://doi.org/10.1093/jaoac/92.1.42>.
- Hanno, S. L., T. M. Casey, H. Rojas De Oliveira, and J. P. Boerman. 2025. Assessment of skeletal muscle dynamics and milk production across a 300-day lactation in multiparous dairy cattle. *J. Dairy Sci.* 108:1092–1102. <https://doi.org/10.3168/jds.2024-25197>.
- Hoh, J. F. Y. 2023. Developmental, physiologic and phylogenetic perspectives on the expression and regulation of myosin heavy chains in mammalian skeletal muscles. *J. Comp. Physiol. B* 193:355–382. <https://doi.org/10.1007/s00360-023-01499-0>.
- Mann, S. 2022. Symposium review: The role of adipose tissue in transition dairy cows: Current knowledge and future opportunities. *J. Dairy Sci.* 105:3687–3701. <https://doi.org/10.3168/jds.2021-21215>.
- Mann, S., A. Abuelo, D. V. Nydam, F. A. Leal Yepes, T. R. Overton, and J. J. Wakshlag. 2016. Insulin signaling and skeletal muscle atrophy and autophagy in transition dairy cows either overfed energy or fed a controlled energy diet prepartum. *J. Comp. Physiol. B* 186:513–525. <https://doi.org/10.1007/s00360-016-0969-1>.
- McArt, J. A. A., D. V. Nydam, G. R. Oetzel, T. R. Overton, and P. A. Ospina. 2013. Elevated non-esterified fatty acids and β -hydroxybutyrate and their association with transition dairy cow performance. *Vet. J.* 198:560–570. <https://doi.org/10.1016/j.tvjl.2013.08.011>.
- McCabe, C., A. Suarez-Trujillo, T. Casey, and J. Boerman. 2021. Relative late gestational muscle and adipose thickness reflect the amount of mobilization of these tissues in periparturient dairy cattle. *Animals (Basel)* 11:2157. <https://doi.org/10.3390/ani11082157>.
- McCabe, C. J., and J. P. Boerman. 2020. Invited Review: Quantifying protein mobilization in dairy cows during the transition period. *Appl. Anim. Sci.* 36:389–396. <https://doi.org/10.15232/aas.2019-01929>.

- Megahed, A. A., M. W. H. Hiew, D. Ragland, and P. D. Constable. 2019. Changes in skeletal muscle thickness and echogenicity and plasma creatinine concentration as indicators of protein and intramuscular fat mobilization in periparturient dairy cows. *J. Dairy Sci.* 102:5550–5565. <https://doi.org/10.3168/jds.2018-15063>.
- Picard, B., and M. Gagaoua. 2020. Muscle fiber properties in cattle and their relationships with meat qualities: An overview. *J. Agric. Food Chem.* 68:6021–6039. <https://doi.org/10.1021/acs.jafc.0c02086>.
- Sadri, H., M. H. Ghaffari, and H. Sauerwein. 2023. Invited review: Muscle protein breakdown and its assessment in periparturient dairy cows. *J. Dairy Sci.* 106:822–842. <https://doi.org/10.3168/jds.2022-22068>.
- Schiaffino, S. 2018. Muscle fiber type diversity revealed by anti-myosin heavy chain antibodies. *FEBS J.* 285:3688–3694. <https://doi.org/10.1111/febs.14502>.
- Schiaffino, S., and C. Reggiani. 1985. 1994. Myosin isoforms in mammalian skeletal muscle. *J. Appl. Physiol.* 77:493–501. <https://doi.org/10.1152/jappl.1994.77.2.493>.
- Song, S., C.-H. Ahn, and G.-D. Kim. 2020. Muscle fiber typing in bovine and porcine skeletal muscles using immunofluorescence with monoclonal antibodies specific to myosin heavy chain isoforms. *Food Sci. Anim. Resour.* 40:132–144. <https://doi.org/10.5851/kosfa.2019.e97>.
- van der Drift, S. G. A., M. Houweling, J. T. Schonewille, A. G. M. Tielens, and R. Jorritsma. 2012. Protein and fat mobilization and associations with serum β -hydroxybutyrate concentrations in dairy cows. *J. Dairy Sci.* 95:4911–4920. <https://doi.org/10.3168/jds.2011-4771>.
- Van Soest, P. J., J. B. Robertson, and B. A. Lewis. 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 74:3583–3597. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2).
- Walter, L. L., T. Gärtner, E. Gernand, A. Wehrend, and K. Donat. 2022. Effects of parity and stage of lactation on trend and variability of metabolic markers in dairy cows. *Animals (Basel)* 12:1008. <https://doi.org/10.3390/ani12081008>.
- Wang, Y., D. Zhang, and Y. Liu. 2024. Research progress on the regulating factors of muscle fiber heterogeneity in livestock: A review. *Animals (Basel)* 14:2225. <https://doi.org/10.3390/ani14152225>.
- Wathes, D. C., Z. Cheng, N. Bourne, V. J. Taylor, M. P. Coffey, and S. Brotherstone. 2007. Differences between primiparous and multiparous dairy cows in the inter-relationships between metabolic traits, milk yield and body condition score in the periparturient period. *Domest. Anim. Endocrinol.* 33:203–225. <https://doi.org/10.1016/j.domaniend.2006.05.004>.
- Westhoff, T. A., T. R. Overton, A. F. Ortega, M. E. Van Amburgh, and S. Mann. 2024. Pre- and postpartum metabolizable protein supply: II. Effects on plasma amino acids and markers of tissue mobilization in transition Holstein dairy cows. *J. Dairy Sci.* 107:10900–10912. <https://doi.org/10.3168/jds.2024-25027>.
- Wicks, J., M. Beline, J. F. M. Gomez, S. Luzardo, S. L. Silva, and D. Gerrard. 2019. Muscle energy metabolism, growth, and meat quality in beef cattle. *Agriculture* 9:195. <https://doi.org/10.3390/agriculture9090195>.
- Wiggans, G. R., and G. E. Shook. 1987. A lactation measure of somatic cell count. *J. Dairy Sci.* 70:2666–2672. [https://doi.org/10.3168/jds.S0022-0302\(87\)80337-5](https://doi.org/10.3168/jds.S0022-0302(87)80337-5).
- Wildman, E. E., G. M. Jones, P. E. Wagner, R. L. Boman, H. F. Troutt Jr., and T. N. Lesch. 1982. A dairy cow body condition scoring system and its relationship to selected production characteristics. *J. Dairy Sci.* 65:495–501. [https://doi.org/10.3168/jds.S0022-0302\(82\)82223-6](https://doi.org/10.3168/jds.S0022-0302(82)82223-6).

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